

CHSH-Form Values Above 2 in Fibonacci Anyon Braiding: A Complete Landscape of Braid Words up to Length 12

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We report a complete sequence-by-sequence landscape of CHSH-form values for six Fibonacci anyons in two encodings, covering all words of length $L = 3$ –12 with two generators and $L = 2$ –9 with three. In the two-generator d_{1b} encoding, braiding alone cannot exceed the classical value: the two generators act on different anyons and commute, and all 8190 braid words of length $L = 1$ –12 give $|S| = 2$ exactly on the leakage-free fusion code. A two-qubit circuit built from the same Fibonacci F - and R -matrices, which does not satisfy the braid relations, reaches $|S| = 2.733$ at $L = 12$ with the standard Fibonacci data (96.6% of the Tsirelson bound $2\sqrt{2}$) and $|S| = 2.811$ at the optimum over a phase deformation; this excess is therefore not a braiding effect. Throughout, “CHSH violation” denotes a CHSH value exceeding 2 on the Fibonacci fusion space, which does not factorize into spatially separated Alice/Bob subsystems; the reported values are signatures of nonseparability and single-algebra contextuality, not Bell violations in the Einstein–Podolsky–Rosen sense. In the three-generator encoding, the generator σ_3 that acts across the bipartition is the only source of entanglement: without it, the state stays in the fusion code as a product state, for all sequences and deformation phases. Under four-dimensional computational-subspace projection the maximum is $|S| = 2.824$ at $L = 8$, and 91.8% of the words exceed 2 at $L = 9$, including 510 words without σ_3 that do so only through the projection; a Horodecki-type score on the native five-dimensional observables exceeds 2 for 42.8%. Sequential braiding on a shared fusion space produces a positive lag-1 autocorrelation of the simulated CHSH value, $\text{ACF}(1) = +0.35$ at 11.22σ , because the state is carried from step to step; a null test with independently drawn phases shows none.

I. INTRODUCTION

Non-Abelian anyons in two-dimensional topological phases support braiding operations that can generate entanglement between logical qubits encoded in the fusion space [3, 4]. For Fibonacci anyons—the simplest non-Abelian species with a universal gate set—explicit braiding sequences implementing quantum gates have been compiled [5, 6], and CHSH-violating settings on the Fibonacci fusion space have been constructed [15]. However, a basic question has remained open: *what fraction of all possible braiding sequences produces CHSH violation, and what controls the boundary between violating and nonviolating sequences?*

We address this question by exhaustive enumeration. For six Fibonacci anyons on a linear fusion tree, we compute the maximal CHSH value $|S|$ for every braiding sequence in two distinct encodings: a two-generator encoding (d_{1b} , $L = 3$ –12, native four-dimensional Hilbert space) and a three-generator encoding ($L = 2$ –8 fully enumerated, $L = 9$ for CHSH-fraction statistics, and the $\delta = 0$ maxima over all 3^L words at $L = 10$ and 12; full five-dimensional fusion space). Throughout, the two-generator case is enumerated in its *circuit model*: a pair of two-qubit gates built from the Fibonacci F - and R -matrices, which do not satisfy the braid relations. We distinguish it from the d_{1b} *fusion code*, the leakage-free five-dimensional realization of the same encoding, in

which both generators act strictly locally. Every $|S|$ value reported for d_{1b} below refers to the circuit model. The enumeration comprises 2^L sequences in d_{1b} and 3^L sequences in the three-generator case (up to $3^{12} = 531\,441$ for the $\delta = 0$ maxima), totaling over 10^6 evaluated unitaries.

Three principal findings emerge:

(i) *Braiding alone does not entangle the two logical qubits of the two-generator encoding.* Its generators act on different anyons and commute; on the leakage-free fusion code all 8190 braid words of length $L = 1$ –12 give $|S| = 2$ exactly, while the two-qubit circuit model built from the same F - and R -matrices reaches $|S| = 2.733$ with the standard Fibonacci data and $|S| = 2.811$ at the sector-phase optimum (Sec. IID).

(ii) *A single generator switches entanglement on in the three-generator encoding.* Without the generator σ_3 that acts across the bipartition, the state stays in the fusion code, where the remaining generators act locally, and the concurrence is $C = 0$ exactly (with $|11\rangle$ assigned to $|4\rangle$, Sec. IV), for all sequences and all values of δ . Over the full gate set, CHSH-form values above 2 are common: under computational-subspace projection 91.8% of the sequences exceed 2 at $L = 9$ (including 510 words without σ_3 , which exceed 2 only through the projection), and a Horodecki-type score on the native five-dimensional observables exceeds 2 for 42.8%, with a further 28.5% exactly at the classical value.

(iii) *Sequential braiding produces temporal correlations in the simulated CHSH value.* When braiding operations are applied sequentially to a shared anyon register, the

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resulting CHSH-value time series exhibits positive autocorrelation ($\text{ACF}(1) = +0.35$, significance 11.22σ), while a null test with independent phases δ yields $\text{ACF} \approx 0$.

These results complement the mutual-information study between topological sectors and CHSH values reported in Ref. [1], which found that classical lattice models ($\mathbb{Z}_2$ gauge theory, XY model) exhibit finite-size correlations $I(T; S) > 0$ between sector labels and Bell-type observables (a finite-size effect that a scaling analysis indicates decays toward zero with system size).

The present work differs from prior studies of Fibonacci braiding in scope and focus. Burke *et al.* [9] characterized the Weyl-chamber structure of Fibonacci braiding gates but did not evaluate CHSH violation. Cui *et al.* [8] established that exact leakage-free entangling gates may not exist for Fibonacci anyons, motivating our two-protocol analysis. Yang *et al.* [10] studied information scrambling through braiding but not CHSH correlations. Experimental demonstrations of non-Abelian statistics [11] have confirmed braiding phases but have not mapped the CHSH landscape. While prior work by Brennen *et al.* [15] established Bell-test constructions for Fibonacci/Ising anyons, Chang, Chu, and Ma [16] studied Bell's inequality in topologically ordered qubit models (Wen-plaquette), and Xu, Ye, and You [20] have independently analyzed three-copy CHSH correlations on Fibonacci anyonic states with a fusion-sector observables construction, the present paper contributes a complete sequence-by-sequence enumeration of CHSH-form values over all braid words of given length in two encodings. Earlier qualitative Bell tests on Majorana-based topological platforms appear in Drummond *et al.* [17] and Romito and Gefen [18], with the Clifford-only limitation of Ising braiding analyzed by Clarke, Sau, and Das Sarma [19].

The paper is organized as follows. Section II defines the two encodings and the CHSH evaluation method. Section III presents the enumeration results and reconciles the two encodings. Section IV establishes the σ_3 -blocking result. Section V analyzes the measurement-protocol dependence of CHSH fractions. Section VI presents the sequential-braiding autocorrelation. Section VII discusses implications and limitations.

II. FRAMEWORK

A. Fusion space of six Fibonacci anyons

We consider six Fibonacci anyons $\tau_1, \dots, \tau_6$ arranged on a linear fusion tree with total charge constrained to the vacuum sector. The Fibonacci fusion rule $\tau \times \tau = \mathbf{1} \oplus \tau$ gives a five-dimensional fusion space spanned by the intermediate charge labels (x_1, x_2, x_3) , where x_k is the total charge of the first $k + 1$ anyons:

$$\begin{aligned} |0\rangle &= (1, \tau, 1), & |1\rangle &= (1, \tau, \tau), \\ |2\rangle &= (\tau, 1, \tau), & |3\rangle &= (\tau, \tau, 1), \\ |4\rangle &= (\tau, \tau, \tau). \end{aligned} \quad (1)$$

The fourth intermediate label $x_4 = d$ is fixed to τ by the vacuum constraint.

B. F- and R-matrices

The Fibonacci F -matrix in the $(\mathbf{1}, \tau)$ basis is

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad (2)$$

satisfying $F^2 = I$ to machine precision ($\Delta < 10^{-17}$). The R -eigenvalues for the two fusion channels are

$$R_{\mathbf{1}} = e^{-4\pi i/5}, \quad R_{\tau} = e^{+3\pi i/5}. \quad (3)$$

C. Braid generators

The five braid generators $\sigma_1, \dots, \sigma_5$ act as 5×5 unitary matrices on the fusion-space basis (1). Each σ_k exchanges anyons τ_k and τ_{k+1} via the sequence $F^{-1} \cdot \text{diag}(R_{\mathbf{1}}, R_{\tau}) \cdot F$ on the relevant two-dimensional subspace, with the remaining basis states acquiring diagonal R -phases. Explicit matrix forms are given in Appendix A.

All four Yang-Baxter relations $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ are satisfied with residuals $\Delta \leq 3.3 \times 10^{-16}$, and distant generators commute ($\Delta \leq 1.5 \times 10^{-17}$ for $|i - j| \geq 2$) [1].

D. Two encodings

We study two encodings of logical qubits into the fusion space:

d_{1b} encoding (two generators). The logical qubits are $Q_A = x_1$ (Alice) and $Q_B = x_3$ (Bob). In the *fusion code*, the four basis states with $x_2 = \tau$ ($|0\rangle, |1\rangle, |3\rangle, |4\rangle$ of Eq. (1)) carry the logical basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$; the fifth state $|2\rangle = (\tau, 1, \tau)$ shares $(x_1, x_3) = (\tau, \tau)$ with $|4\rangle$ and is not reached without σ_3 . The available braid generators are σ_2 (braids $\tau_2\tau_3$, acts on Alice) and σ_4 (braids $\tau_4\tau_5$, acts on Bob). They act on different anyons and commute, so braiding alone cannot entangle Alice and Bob: over all 8190 braid words of length $L = 1$ –12 the concurrence stays at machine zero and $|S| = 2$ exactly (`d9_fusion_code_braiding_only.py`). The *circuit model* used for the d_{1b} values below is a two-qubit circuit whose two gates are built from the Fibonacci F - and R -matrices: one acts on Alice's qubit alone, the other acts on Bob's qubit conditioned on Alice's; the two gates do not satisfy the braid relations. We denote them AM and MB; they do not commute ($\|[\text{AM}, \text{MB}]\|_{\infty} \approx 0.30$).

A phase δ , called the sector phase in this paper, is introduced by modifying $R_{\tau} \rightarrow R_{\tau} \cdot e^{i\delta}$. It is a deformation parameter of the representation, not a physical topological sector: for generic $\delta \neq 0$ the braid relations fail, and only $\delta = 0$ is the standard Fibonacci theory (App. C).

Three-generator encoding (full 5D). All five basis states (1) are retained, and the gate set is $\{\sigma_2, \sigma_3, \sigma_4\}$. Here σ_2 acts locally on Alice (x_1), σ_4 acts locally on Bob (x_3), and σ_3 directly crosses the bipartition by refusing x_2 . This provides the standard structure of single-qubit rotations plus entangling gate for universal quantum computation [5].

E. Interpretation: nonseparability and contextuality

Throughout this paper, “CHSH violation” refers to CHSH values exceeding 2 computed on the Fibonacci anyon fusion space. This fusion space does not admit a tensor-product factorization into spatially separated Alice/Bob subsystems. Accordingly, the reported values are signatures of *nonseparability* on a nonfactorized fusion space, not Bell violations in the Einstein–Podolsky–Rosen sense.

On the full five-dimensional fusion space, the Protocol B observables commute exactly across the bipartition ($[A_i, B_j] = 0$ for all i, j , Sec. V), so Tsirelson’s bound remains the relevant ceiling and is respected throughout (at most $|S| = 2.824$, 99.84% of $2\sqrt{2}$, under the four-dimensional projection of Protocol A). We therefore classify the reported correlations as a form of *single-algebra contextuality* [24] rather than EPR-type nonlocality. In algebraic quantum field theory, maximal Bell-CHSH violation is generic for observables localized in spacelike-separated regions of a local net of von Neumann algebras [40]; we cite this as context for correlations defined on a single non-Abelian algebra, and note that the present setting does not exhibit the algebraic locality structure those results assume. Genericity statements of this kind also have a measure-theoretic counterpart for anyonic systems: there, the states with vanishing logarithmic negativity form a set of measure zero, in contrast with ordinary qubit systems, and this is proved for multiplicity-free categories [25]; we cite this as context and note that the quantity there is entanglement, which is necessary but not sufficient for CHSH violation, and that it is a statement about a measure on a continuous state space, whereas the genericity reported here is a counted fraction over a discrete enumeration of braiding sequences. Related studies of Bell-operator representations on single quantum systems are reported in Refs. [23, 39]. This reading is consistent with the anyonic-entanglement structure described in Ref. [26], and with the broader point—sharpened by the resolution of the Tsirelson problem [41]—that the appropriate correlation bound depends on the algebraic separation assumed between the measurement structures, not merely on the numerical value of $|S|$. We are not aware of a sharp no-go theorem covering this specific setting; we make no prohibition claim here and instead report the values as nonseparable correlations bounded by, but not violating, Tsirelson’s inequality.

F. CHSH evaluation

For each braiding sequence $\mathcal{B}$ and sector phase δ , we compute the unitary $U(\mathcal{B}, \delta)$ and the output state $|\psi\rangle = U|0\rangle$. The maximal CHSH value is obtained via the Horodecki criterion [7]: for a pure two-qubit state with concurrence C ,

$$|S|_{\max} = 2\sqrt{1 + C^2}. \quad (4)$$

This bound is tight and achieved by optimal measurement angles derived from the singular-value decomposition of the correlation matrix $T_{ij} = \text{Re} \langle \psi | A_i \otimes B_j | \psi \rangle$.

In the d_{1b} circuit model, C is the concurrence of the two-qubit state of the circuit, and Eq. (4) gives an encoding-level two-qubit value; it is not a statement about spatially separated subsystems (Sec. II E). In the three-generator encoding, we evaluate CHSH under two protocols described in Sec. V.

For each length L , the CHSH-violation fraction is defined as $\text{CHSH}\% = |\{(\mathcal{B}, \delta) : |S| > 2\}| / |\{(\mathcal{B}, \delta)\}|$.

III. RECONCILIATION OF TWO ENCODINGS

A. d_{1b} : Two-generator landscape

Table I presents the optimal CHSH value, concurrence, and winning sequence for each braid length $L = 3$ –12 in the d_{1b} encoding, obtained by exhaustive enumeration over all 2^L sequences; the per-sequence sector-phase optima are reported on the 50-point grid of the deposited `s_delta_sweeps.json`. (At $L = 2$, a single gate yields $|S| = 2.097$, $C = 0.315$.)

TABLE I. Optimal CHSH-form values in the d_{1b} circuit model (two gates AM, MB), maximized over the 50-point sector-phase grid; sequences are written with A = AM and B = MB. $|S|$ is computed via the Horodecki bound from the maximal concurrence C . At $\delta = 0$, with the standard Fibonacci data, $L = 12$ gives $|S| = 2.733$ ($C = 0.932$, sequence ABBAABAB-BAAB; Ref. [1]).

L	$ S $	C	% Tsirelson	Sequence
3	2.199	0.457	77.7	ABB
4	2.225	0.488	78.7	ABAB
5	2.348	0.615	83.0	AABAB
6	2.433	0.693	86.0	AAABAB
7	2.536	0.780	89.7	ABBAABB
8	2.540	0.783	89.8	BABBABAB
9	2.639	0.861	93.3	ABBBBABAB
10	2.696	0.904	95.3	ABBAABBAAB
11	2.784	0.968	98.4	ABABBBBABAB
12	2.811	0.988	99.4	ABABABABABAB

The concurrence increases monotonically from $C = 0.457$ at $L = 3$ to $C = 0.988$ at $L = 12$, with no sat-

uration plateau. The optimal sequence at $L = 12$, the alternating commutator structure $(AM \cdot MB)^6$, achieves 99.4% of the Tsirelson bound at the sector-phase optimum; at $\delta = 0$ the $L = 12$ maximum is 96.6%.

We note that an earlier analysis [1] reported $|S| = 2.554$ and $C = 0.903$ at $L = 12$ for this encoding. That value was an optimization artifact: the original code employed six fixed angular starting points and a 9^4 grid for CHSH angle optimization, which missed the global maximum. Our Horodecki-based evaluation, which requires only the concurrence (a basis-independent quantity), recovers the true optimum.

B. Three-generator landscape

Table II presents the analogous results for the three-generator encoding, enumerated over 3^L sequences for $L = 2-8$.

Under the Protocol A projection the landscape approaches the Tsirelson bound already at $L = 8$, reaching $|S| = 2.824$ ($C = 0.997$, 99.84% of the Tsirelson bound) at the enumeration maximum; exhaustive search over all $3^8 = 6561$ sequences finds no single braid word that saturates the bound exactly. The maximizing word gives $|S| \approx 2.59$ ($C = 0.824$) if $|11\rangle$ is assigned to $|4\rangle$ alone, so part of the value comes from the projection (Sec. V). The full sector-phase dependence $|S|(\delta)$ of the first $\delta = 0$ maximizers at $L = 3, 8, 12$ is shown in Fig. 1; Fig. 2 places it beside the two-generator landscape over all deposited lengths.

At $L = 2$, no sequence produces entanglement in the assignment $|11\rangle \leftarrow |4\rangle$ ($C = 0$): words without σ_3 consist of gates local to Alice or Bob, and in words with σ_3 the generator acts on a state supported on basis states that differ in one qubit only, where it acts as a phase. The Protocol A value 2.029 of the word $\sigma_2\sigma_4$ (Table II) comes from the projection alone. From $L = 3$ onward, the Protocol A concurrence exceeds 0.92 (rising to 0.997 at $L = 8$), reflecting the availability of σ_3 as a primitive entangling gate.

The CHSH-violation fraction under 4D projection rises monotonically: 44% at $L = 3$, 79% at $L = 6$, 92% at $L = 9$ (see Sec. V for the protocol dependence of these fractions).

C. Comparing the two encodings

The two encodings differ in the reachable group: the circuit gates $\{AM, MB\}$ generate a six-dimensional proper subgroup of $SU(4)$ (entangling but not universal), whereas $\{\sigma_2, \sigma_3, \sigma_4\}$ acts on the five-dimensional fusion space as the standard B_4 braid representation, which is dense in $SU(2) \times SU(3)$ on the two sectors of the intermediate charge x_2 [5]; the four-dimensional computational-subspace measurement is a projection of this action, not

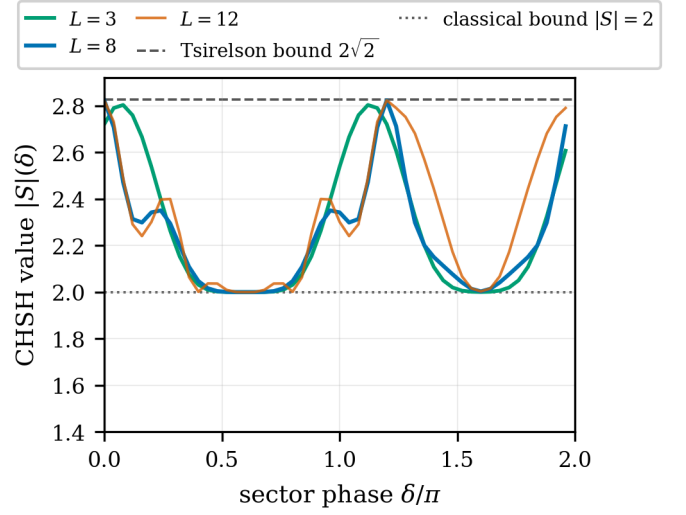


FIG. 1. Sector-phase dependence $|S|(\delta)$ of the three-generator sequences at braid lengths $L = 3, 8, 12$ that maximize $|S|$ at $\delta = 0$ under Protocol A (first maximizer in lexicographic order, from the enumeration of all 3^L words). The CHSH value ranges from the classical bound $|S| = 2$ in the “dead” sector phases up to $|S| = 2.824$ (99.84% of the Tsirelson bound $2\sqrt{2}$, dashed) at $L = 8$ and 12 , where both curves reach this maximum at $\delta = 0$ and $\delta = 6\pi/5$ and lie on top of each other there; at $L = 3$ the maximum is 2.804. The curves at $L = 8$ and 12 carry more structure than at $L = 3$ (Fourier fits with four harmonics: $R^2 = 0.997, 0.837, 0.914$; App. B). Lines connect the $N = 50$ computed sector-phase points, so structure finer than the sampling is not resolved. Data: `s_delta_sweeps.json` (concept-DOI 10.5281/zenodo.19601352).

an invariant subspace. They differ further in the word length required to approximate a given unitary.

In the three-generator encoding, σ_3 acts across the Alice–Bob bipartition and provides the entangling step of the standard *rotate–entangle–rotate* decomposition. In the d_{1b} circuit model, the entangling step is the conditional gate MB, which is not a braid; braiding with σ_2 and σ_4 alone does not entangle (Sec. IID). The representation-dependence of the CHSH operator norm on higher-dimensional Hilbert spaces is discussed in Ref. [21].

IV. THE CROSS-BIPARTITION GENERATOR

σ_3

We now show that, in the three-generator encoding, entanglement between the logical qubits requires the generator σ_3 .

TABLE II. Optimal CHSH-form values in the three-generator encoding ($\sigma_2, \sigma_3, \sigma_4$) at $\delta = 0$ under 4D computational-subspace projection (Protocol A, Sec. V). The $|S|$ column is the exhaustive-enumeration maximum $S_{\max}^{4D}$ from the canonical `landscape_validation.json`; $C = \sqrt{(|S|/2)^2 - 1}$; % Tsirelson = $|S|/2\sqrt{2}$. The $L = 2$ entry (`d10_protocol_a_projection_audit.py`) exceeds 2 only through the projection (Sec. V).

L	$ S $	C	% Tsirelson	Sequence
2	2.029	0.171	71.74	(projection only)
3	2.722	0.923	96.22	(enumeration max)
4	2.722	0.923	96.22	(enumeration max)
5	2.791	0.973	98.68	(enumeration max)
6	2.791	0.973	98.68	(enumeration max)
7	2.791	0.973	98.68	(enumeration max)
8	2.824	0.997	99.84	(enumeration max)

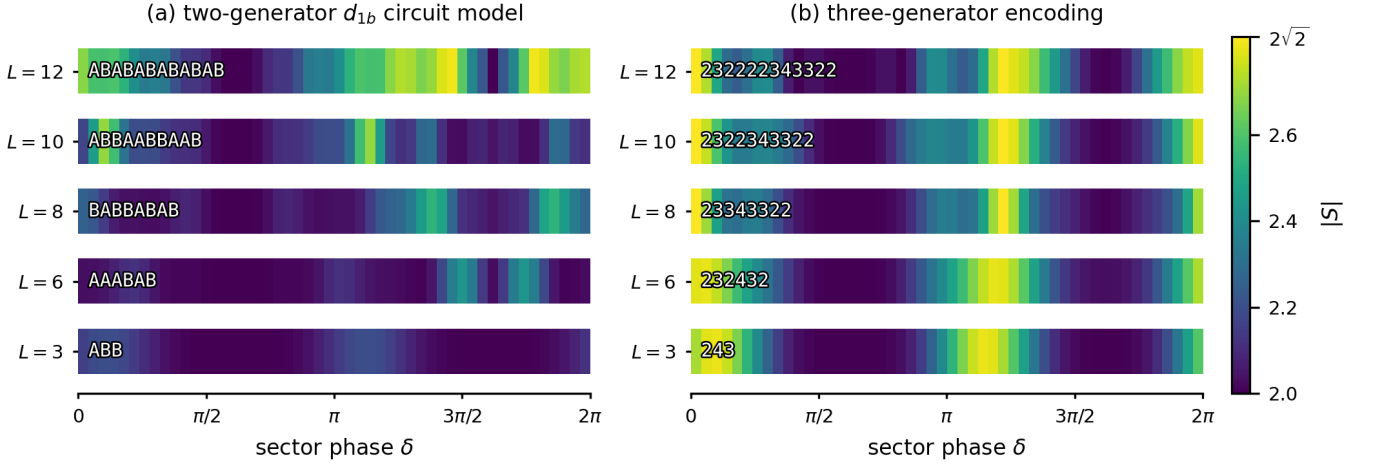


FIG. 2. CHSH value $|S|$ of the sequence listed for each length (Table I for (a); for (b) the first $\delta = 0$ maximizer under Protocol A) as a function of the sector phase δ , for the two-generator d_{1b} circuit model (a) and the three-generator five-dimensional encoding (b). Each strip is one braid length; L is a categorical axis, since the deposited sweeps cover $L \in \{3, 6, 8, 10, 12\}$ and nothing is drawn between them. Both panels share one color scale, from the classical bound $|S| = 2$ to the Tsirelson bound $2\sqrt{2}$. Data: `s_delta_sweeps.json` (concept-DOI 10.5281/zenodo.19601352).

A. Blocking the cross-bipartition generator

In the three-generator encoding, we replace σ_3^1 by the 5×5 identity matrix and evaluate the concurrence, with $|11\rangle$ assigned to $|4\rangle$, on three representative $\{\sigma_2, \sigma_4\}$ configurations, each swept over 50 linearly spaced sector-phase values $\delta \in [0, 2\pi)$ (150 evaluations in total).

The result is absolute:

$$C(\mathcal{B}, \delta)|_{\sigma_3 \rightarrow I} = 0 \quad \forall \mathcal{B}, \forall \delta. \quad (5)$$

Across all tested blocked configurations (three representative sequences at $L = 3, 6, 8$ plus the d_{1b} winner at

$L = 12$ with each of its two generators blocked in turn, each configuration swept over 50 sector phases, totaling $5 \times 50 = 250$ evaluations), the maximal concurrence is $C < 4.4 \times 10^{-16}$ —machine zero. Figure 3 contrasts the two cases for a representative $L = 8$ sequence $\sigma_2\sigma_3\sigma_4^4\sigma_3\sigma_2$: with σ_3 active (Protocol A), $|S|(\delta)$ rises above the classical bound, up to $|S| = 2.808$ at $\delta/\pi = 1.24$ and 1.96 ; with σ_3 blocked, it is pinned to the classical bound for every δ in the assignment $|11\rangle \leftarrow |4\rangle$, and reaches at most $|S| = 2.024$ under the Protocol A projection, through the projection alone.

B. Physical mechanism

The vanishing of entanglement under σ_3 -blocking has a transparent algebraic origin: σ_2 preserves the x_3 label (Bob's qubit) and σ_4 preserves the x_1 label (Alice's qubit). Without σ_3 , which re-fuses the intermediate

¹ Index convention across the companion series: with $n = 6$ anyons and the bipartition cut after anyon 3, the cross-bipartition generator of this paper is σ_3 , while σ_4 acts locally on Bob's side. In the companion paper [2] ($n = 8$ anyons, cut after anyon 4) the cross-bipartition role is played by its σ_4 ; the index shift is purely notational.

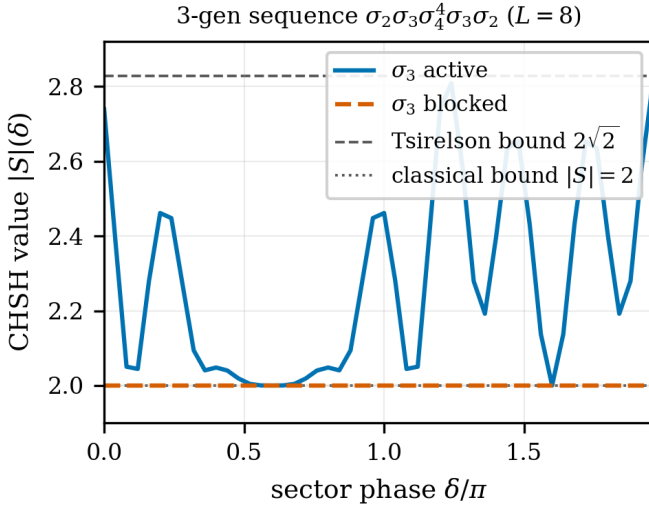


FIG. 3. The σ_3 generator as a switch for entanglement, for a representative $L = 8$ three-generator sequence $\sigma_2\sigma_3\sigma_4^4\sigma_3\sigma_2$. With σ_3 active (solid, Protocol A), $|S|(\delta)$ rises above the classical bound, up to $|S| = 2.808$ at $\delta/\pi = 1.24$ and 1.96 ; replacing σ_3 by the identity (dashed, assignment $|11\rangle \leftarrow |4\rangle$) makes the concurrence vanish identically [Eq. (5)], pinning $|S|$ to the classical bound $|S| = 2$ for every δ . Under the Protocol A projection the blocked sequence reaches at most $|S| = 2.024$, through the projection alone. Data: `topology_switch_results.json` (concept-DOI 10.5281/zenodo.19601352).

charge x_2 and thereby couples the two branches of the fusion tree, the generators act independently on the two logical qubits. The resulting state is necessarily a product state, regardless of sequence length or sector phase. This holds in the assignment $|11\rangle \leftarrow |4\rangle$, in which σ_2 and σ_4 act as single-qubit gates; the four-dimensional projection of Protocol A (Sec. V) instead assigns $|11\rangle \leftarrow (|2\rangle + |4\rangle)/\sqrt{2}$, keeps only half of the weight of $|4\rangle$, and can therefore report $C > 0$ for such product states.

C. The two-generator circuit model

In the d_{1b} circuit model, which has no σ_3 , blocking either AM or MB also collapses C to zero exactly for the optimal sequence $(\text{AM} \cdot \text{MB})^6$ at all 50 sector phases, for a different reason: AM alone acts on Alice only, while MB alone changes Bob’s qubit conditioned on Alice’s, which stays in $|0\rangle$ when AM is removed. Entanglement in the circuit model therefore requires both gates.

D. Interpretation

In the three-generator encoding the σ_3 switch shows which braiding operations make entanglement accessible: from the product input state, only a generator acting across the bipartition can create it, as in any quantum

circuit acting on a product input. Section VI returns to this result in a time-resolved setting.

V. MEASUREMENT PROTOCOL AND COMPUTATIONAL-SUBSPACE LEAKAGE

The CHSH-violation fractions reported in Sec. III for the three-generator encoding depend on how observables are defined in the five-dimensional fusion space. We distinguish two measurement protocols and present results for both.

A. Protocol A: Computational-subspace projection

The four-dimensional computational subspace is defined by identifying the logical two-qubit basis as $|00\rangle \leftarrow |0\rangle$, $|01\rangle \leftarrow |1\rangle$, $|10\rangle \leftarrow |3\rangle$, and $|11\rangle \leftarrow (|2\rangle + |4\rangle)/\sqrt{2}$. The last mapping coherently recombines the two basis states with $x_1 = \tau$, $x_3 = \tau$ that differ only in the intermediate charge x_2 . Because a state without weight on $|2\rangle$ keeps only half of its $|4\rangle$ weight under this mapping, the projection can turn a product state into a non-product one: at $L = 9$, 510 of the 18064 words that exceed $|S| = 2$ under Protocol A contain no σ_3 and exceed 2 only through the projection (`d10_protocol_a_projection_audit.py`). After projection and renormalization, the standard two-qubit Horodecki criterion (4) applies.

Under Protocol A, the CHSH-violation fraction rises monotonically with braid length and reaches 91.8% at $L = 9$ (Table III). The maximal CHSH value under Protocol A saturates at $|S| = 2.824$ (Table II), within 0.16% of the Tsirelson bound $|S| = 2\sqrt{2}$ —approached but, per the exhaustive 3^8 -sequence search, not exactly attained by any single braid word at $L = 8$, reflecting the residual cost of the four-dimensional computational-subspace projection rather than a different physical limit.

B. Protocol B: Native 5D CHSH

We define Alice and Bob observables directly in the five-dimensional fusion space:

$$Z_A = \text{diag}(+1, +1, -1, -1, -1), \quad (6)$$

$$Z_B = \text{diag}(+1, -1, -1, +1, -1), \quad (7)$$

assigning eigenvalues ± 1 according to the x_1 and x_3 labels. The X -observables implement spin flips within each qubit sector: X_A swaps $|0\rangle \leftrightarrow |3\rangle$ and $|1\rangle \leftrightarrow |4\rangle$, while X_B swaps $|0\rangle \leftrightarrow |1\rangle$ and $|3\rangle \leftrightarrow |4\rangle$. The state $|2\rangle = (\tau, 1, \tau)$ has no flip partner in either qubit sector, giving $X|2\rangle = 0$ —i.e., the X and Y observables have spectrum $\{-1, 0, +1\}$ (trit-valued).

We apply the Horodecki formula to the 3×3 correlation matrix $T_{ij} = \text{Re} \langle \psi | A_i \otimes B_j | \psi \rangle$, yielding $|S|_{\max} =$

$2\sqrt{m_1 + m_2}$ where m_1, m_2 are the two largest eigenvalues of $T^T T$. Because the formula assumes ± 1 -valued observables on a two-qubit tensor product, which these trit-valued observables on a single five-dimensional space are not, we report the result as a Horodecki-type score rather than as a Bell test. All observables commute across the bipartition ($[A_i, B_j] = 0$ for all i, j).

C. Protocol comparison

Table III compares the two protocols across braid lengths $L = 3$ –9 (exhaustive enumeration of all 3^L sequences).

Figure 4 summarizes both protocols across braid length. The two protocols agree in direction—under both, the violating fraction rises monotonically with L —but not in magnitude: Protocol A exceeds one half from $L = 4$ onward, whereas Protocol B reaches 42.8% at $L = 9$. They differ in two further respects:

(i) The maximal CHSH value in Protocol B saturates at $|S|_{\max}^{5D} \approx 2.517$ from $L = 5$ onward, while Protocol A approaches the Tsirelson bound (within 0.16%). The gap reflects the zero eigenvalue of the X and Y observables on $|2\rangle$: a maximally entangled singlet-like state cannot be fully realized in the five-dimensional space using trit-valued observables.

(ii) The CHSH fraction in Protocol B rises monotonically to 42.8% at $L = 9$, remaining below one half at every length, while under Protocol A it exceeds one half from $L = 4$ onward.

D. Leakage statistics

The computational-subspace fidelity is the squared overlap onto the orthonormal logical basis $\{|0\rangle, |1\rangle, |3\rangle, (|2\rangle + |4\rangle)/\sqrt{2}\}$,

$$N^2 = |\psi_0|^2 + |\psi_1|^2 + |\psi_3|^2 + \frac{1}{2}|\psi_2 + \psi_4|^2, \quad (8)$$

which measures the fraction of the five-dimensional state that survives projection into the two-qubit subspace and satisfies $N^2 \in [0, 1]$ by construction. At $L = 9$, the minimum fidelity across all 19 683 sequences is $N_{\min}^2 = 0.131$, meaning that for some sequences up to 86.9% of the amplitude resides outside the computational subspace.

This is consistent with the expectation that exact leakage-free entangling gates may not exist for Fibonacci anyons [8]. Protocol A implicitly post-selects on the computational subspace by renormalization; Protocol B avoids this at the cost of a reduced maximal $|S|$.

E. Summary statement

CHSH-form values above 2 are not confined to special braiding sequences under either evaluation. Their frequency and size, however, depend on the evaluation: the

CHSH fraction (42.8% vs. 91.8% at $L = 9$) and the saturation value of $|S|_{\max}$ (2.517 vs. 2.824) both differ, reflecting the incomplete flip symmetry of the five-dimensional fusion space and, for Protocol A, the contribution of the projection itself.

Note on terminology. In this paper, “Protocol B” refers specifically to the bipartite native-5D CHSH evaluation with trit-valued Z and X -like observables on the full five-dimensional fusion space (Sec. V, this paper). Other companion analyses that probe alternative non-bipartite CHSH constructions on the same fusion space (e.g., minimal three-diagonal-parity single-particle constructions) are distinct in scope and are not covered by the present landscape; the bipartite Protocol B used here remains well-defined as a Horodecki-type score in its own right.

We emphasize that the d_{1b} circuit model (Sec. III, Table I) acts on a four-dimensional space with no leakage and yields the sector-phase grid maximum $|S| = 2.811$ of the d_{1b} circuit model ($\delta/\pi = 1.44$ and 1.76); at $\delta = 0$, with the standard Fibonacci data, the $L = 12$ maximum is $|S| = 2.733$, the value adopted as the Fibonacci reference in Ref. [1].

VI. SEQUENTIAL BRAIDING AND TEMPORAL CORRELATIONS

We now show that sequential braiding on a shared fusion space produces temporal correlations in the simulated CHSH-value series.

A. Setup

A register of six Fibonacci anyons is initialized in the vacuum state $|0\rangle$ and subjected to $N = 1000$ sequential braiding steps. At each step n , a generator $h_n \in \{\sigma_2, \sigma_3, \sigma_4\}$ is drawn from a specified distribution, the state is updated as $|\psi_n\rangle = h_n|\psi_{n-1}\rangle$, and its CHSH value under the four-dimensional projection (Protocol A, Sec. V), $S_n = S[\psi_n]$, is recorded. This is repeated for $M = 20$ independent runs.

Three models are compared:

Model A (i.i.d.): each generator is chosen uniformly at random ($p = 1/3$ each).

Model B (Markov-persistent): with probability 0.6, the previous generator is repeated; otherwise a uniform draw from the remaining two.

Model C (no σ_3): generators drawn from $\{\sigma_2, \sigma_4\}$ only ($p = 1/2$ each), serving as a control.

B. Pillar 2: Null test

As a prerequisite, we verify that *independent* sector phases produce no autocorrelation. Drawing $N = 10\,000$

TABLE III. CHSH-violation statistics under two measurement protocols. $|S|_{\max}$ is the largest CHSH value across all sequences. CHSH% counts sequences with $|S| > 2 + \tau$, $\tau = 10^{-10}$; the “at bound” column counts those with $||S| - 2| \leq \tau$, which are never assigned to either side. $N_{\min}^2$ is the minimum computational-subspace fidelity. Columns may not sum to 100 due to rounding.

L	$ S _{\max}^{5D}$	$ S _{\max}^{4D}$	CHSH% _{5D}	at bound _{5D}	CHSH% _{4D}	Δ	$N_{\min}^2$
3	2.305	2.722	7.4	92.6	44.4	+37.0	0.736
4	2.305	2.722	9.9	80.2	61.7	+51.9	0.635
5	2.517	2.791	16.5	66.3	72.4	+56.0	0.480
6	2.517	2.791	23.6	53.4	79.0	+55.4	0.281
7	2.517	2.791	33.7	43.5	84.2	+50.5	0.281
8	2.517	2.824	39.1	35.6	88.5	+49.4	0.183
9	2.517	2.824	42.8	28.5	91.8	+49.0	0.131

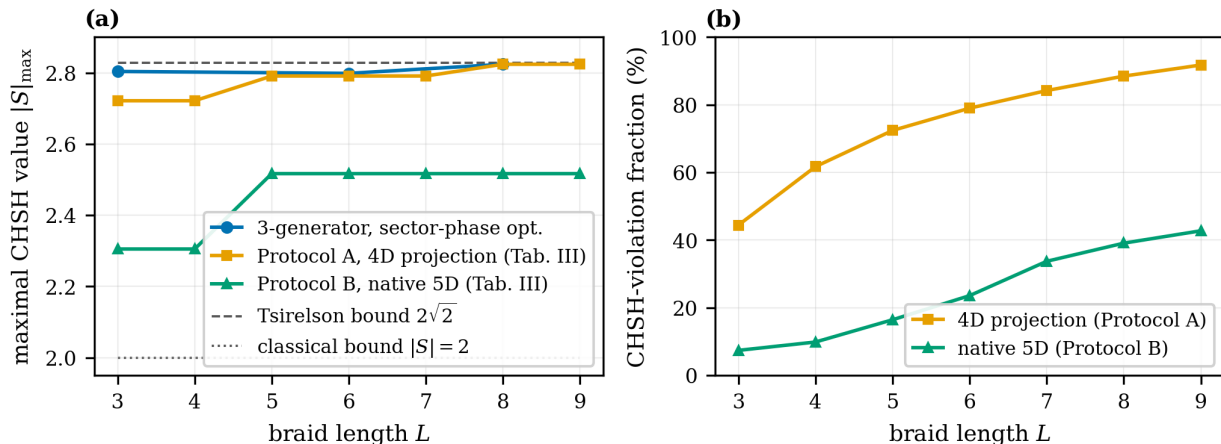


FIG. 4. Braiding-length dependence of the CHSH landscape. (a) Maximal CHSH value $|S|_{\max}$ versus braid length L for the three-generator encoding at the sector-phase optimum of the first $\delta = 0$ maximizers under Protocol A for $L = 3, 6, 8$ (per- L maxima of `s_delta_sweeps.json`: 2.804, 2.798, 2.824; at $L = 8$ this point coincides with the Protocol A value and lies beneath its marker), and for the two operational measurement protocols (Table III): Protocol A (four-dimensional computational-subspace projection, approaching $2\sqrt{2}$ within 0.16%) and Protocol B (native five-dimensional, trit-valued observables, saturating near 2.517). Dashed: Tsirelson bound $2\sqrt{2}$; dotted: classical bound $|S| = 2$. (b) Fraction of braiding sequences with $|S| > 2 + \tau$ under the two protocols. Both rise monotonically with L , reaching 91.8% (Protocol A) and 42.8% (Protocol B) at $L = 9$; under Protocol B a further 28.5% of sequences lie exactly at the classical bound. Data: `landscape_validation.json` and `s_delta_sweeps.json` (concept-DOI 10.5281/zenodo.19601352).

sector phases δ_i uniformly from $[0, 2\pi)$ and computing the CHSH value for each, the lag-1 autocorrelation is $\text{ACF}(1) = +0.002$ (Model A configuration) and $\text{ACF}(1) = +0.010$ (d1b configuration), both within the 95% confidence interval $\pm 1.96/\sqrt{N} = \pm 0.020$ (seeded evaluation deposited as `d8_acf_null_robustness.py`).

The null hypothesis ($\text{ACF} = 0$ under i.i.d. sectors) is not rejected. This confirms that any nonzero autocorrelation observed in the sequential-braiding models is a genuine dynamical effect, not a statistical artifact.

C. Pillar 3: Sequential-braiding autocorrelation

Table IV presents the measured autocorrelation for the three models.

Both Models A and B exhibit highly significant posi-

TABLE IV. Lag-1 autocorrelation of the CHSH time series under sequential braiding. Significance is measured against the shuffle confidence band (1000 random permutations of the observed series). CHSH values under Protocol A; CHSH% is the fraction of steps with $|S| > 2 + 10^{-10}$.

Model	Generators	ACF(1)	Significance	CHSH%
A (i.i.d.)	$\sigma_2, \sigma_3, \sigma_4$	+0.346	11.22σ	99.5
B (Markov)	$\sigma_2, \sigma_3, \sigma_4$	+0.352	10.53σ	99.0
C (no σ_3)	σ_2, σ_4	+0.035	1.13σ	81.0

tive autocorrelation, with Model B only marginally above Model A (0.352 vs. 0.346). This indicates that the autocorrelation arises primarily from *state continuity*—each braid step applies a finite rotation, keeping $|\psi_n\rangle$ close to $|\psi_{n-1}\rangle$ in Hilbert space—rather than from generator-

level persistence.

Model C shows no significant autocorrelation ($\text{ACF}(1) = +0.035, 1.13\sigma$). Its values are not constant under the four-dimensional projection: the product states reached by σ_2 and σ_4 acquire a nonzero projected concurrence, the same projection effect as for the 510 words without σ_3 (Sec. V), so 81% of its values lie above 2. In the natural assignment ($|11\rangle$ assigned to $|4\rangle$) the concurrence stays zero and $|S| = 2$ exactly, as in Sec. IV.

D. Robustness

Replacing the vacuum initial state with a seeded random normalized five-dimensional vector yields $\text{ACF}(1) = +0.314$ (Model A) and $+0.336$ (Model B), confirming that the effect survives initialization.

The full autocorrelation range across the four deposited setups (vacuum and random initialization, Models A and B) is $\text{ACF}(1) \in [0.31, 0.36]$, with the representative value for long runs (Model A, vacuum start) being $+0.35$.

E. Summary of the three results

The three pillars combine as follows:

Pillar 1 (Sec. IV): σ_3 -blocking shows that, in the three-generator encoding, a single generator switches entanglement on and off, as expected: without it, all gates are local.

Pillar 2 (this section): the null test confirms that i.i.d. sector phases produce no autocorrelation—the method has no intrinsic bias.

Pillar 3 (this section): sequential braiding produces $\text{ACF}(1) = +0.35$ at $> 10\sigma$ —state continuity along the braid sequence produces genuine temporal correlations of the simulated CHSH value.

F. Proposed experimental test

In an experiment with sequential anyon braiding, the detection sequence of CHSH-violation outcomes may exhibit positive autocorrelation. We define the signed per-run CHSH term

$$g_n = (-1)^{a_n b_n} x_n y_n \in \{-1, +1\}, \quad (9)$$

where $a_n, b_n \in \{0, 1\}$ label the measurement settings and $x_n, y_n \in \{-1, +1\}$ the outcomes of run n ; $\text{ACF}(1)$ denotes the lag-1 autocorrelation of the g_n series. For independently re-prepared trials, such as the spacelike-separated trials of loophole-free Bell tests [12, 13, 34], no systematic positive ACF is expected. The simulated value ($\text{ACF}(1) \approx 0.35$) refers to the CHSH value of the state along the braid sequence, evaluated with state-dependent optimal settings and with no measurement

between steps. An experiment records single-run CHSH terms g_n and measures in every run; whether a positive autocorrelation survives measurement back-action, single-shot sampling noise and fixed settings is not computed here, so $\text{ACF} > 0$ for g_n is a hypothesis to be tested on platforms that braid sequentially on a shared fusion register, not a derived prediction.

As a null control, we computed $\text{ACF}(1)$ on the published outcome sequence of the openly available loophole-free test of Hensen *et al.* [12] (data DOI 10.4121/uuid:6e19e9b2-4a2d-40b5-8dd3-a660bf3c0a31, $N = 245$ valid trials), finding $\text{ACF}(1) = -0.047$ (-0.64σ , $|\text{ACF}(1)| < 0.125$ at 2σ)—consistent with zero, as expected: the sequential-braiding mechanism proposed here is structurally absent from independently re-prepared Bell trials, so this is a genuine null control rather than a failed prediction. The earlier test of Giustina *et al.* [13] is closed as a venue for this analysis, as its outcome sequence is available only on request; the higher-statistics Shalm–NIST test [34], although also openly available, was not analyzed here. The appropriate venue to test the sequential-braiding hypothesis is a platform that physically implements braiding sequentially on a shared register, such as the digital Fibonacci-anyon realization of Xu *et al.* [43] or the digital non-Abelian Ising-anyon braiding of Andersen *et al.* [11]; both are gate-based (digital) realizations rather than the analog continuous-braiding transport assumed in the present model, a distinction relevant to any quantitative comparison. The memory loophole in sequential Bell-type data is well characterized [35]; related but distinct analyses of the same loophole-free data sets include a no-signaling test [36] and a Kolmogorov-complexity analysis of the outcome sequences [37, 38], neither of which computes the lag-1 ACF proposed here; improved statistical methodology for Bell-experiment data is discussed in Ref. [44].

VII. DISCUSSION

A. Braiding, circuit, and the crossing generator

The central finding of this work is that braiding alone does not entangle the logical qubits of the two-generator encoding: its generators act on different anyons and commute, and all 8190 braid words of length $L = 1$ –12 give $|S| = 2$ exactly on the fusion code. Entanglement appears once a gate acts across the bipartition—the conditional gate of the d_{1b} circuit model, or the braid generator σ_3 of the three-generator encoding. In the latter, values above 2 are not confined to special sequences: under computational-subspace projection more than half of all sequences exceed 2 for $L \geq 4$ (a count that includes words without σ_3 , which exceed 2 only through the projection), and under the native five-dimensional score the fraction is smaller but rises monotonically with braid length, to 42.8% at $L = 9$, with a further 28.5% exactly at the

classical value.

B. The cross-bipartition generator

In the three-generator encoding, the σ_3 -blocking result shows which braiding operations make entanglement accessible: σ_3 is the only generator acting across the Alice–Bob bipartition, and without it the concurrence stays exactly zero, as it must for gates local to either side acting on a product input (in the assignment $|11\rangle \leftarrow |4\rangle$). The companion analysis of Ref. [1] reports an association between sector labels and CHSH values in classical $\mathbb{Z}_2$ lattice gauge theory, which it finds to be a finite-size effect of nontopological origin.

C. Implications for topological quantum computation

The σ_3 -blocking mechanism has a direct interpretation in the context of topological quantum computing: any braiding protocol that inadvertently excludes the cross-bipartition generator will produce zero entanglement, regardless of circuit depth. The leakage analysis (Sec. V) shows that this concern extends beyond gate selection to the measurement protocol, as up to 86.9% of state weight can reside outside the computational subspace. This reinforces the finding of Cui *et al.* [8] that exact leakage-free entangling gates for Fibonacci anyons may not exist.

D. Limitations

Several limitations qualify our results:

(i) All computations are *numerical simulations* within the idealized Fibonacci TQFT at zero temperature and zero decoherence. Topological protection is expected to suppress braiding-phase noise [3]. For a closed-form two-parameter degradation model of our own, $|S| = 2\lambda^2\sqrt{1+v^4C^2}$ (charge-survival probability λ , dephasing visibility v , ideal concurrence C), pure dephasing ($v < 1$, $\lambda = 1$) bounds $|S|$ from below by the classical value 2 (attained only in the fully-dephased limit $v \rightarrow 0$) and never drives it below—the violation *vanishes* rather than surviving degraded. Only charge-changing quasiparticle poisoning ($\lambda < 1$) can push $|S|$ below 2, at the Werner-state threshold [7] $\lambda^* = 2^{-1/4} = 0.8409$ for an ideal ($C = 1$) Bell state and $\lambda^* = 0.8416$ for the actually braided state ($C = 0.9968$)—a poisoning tolerance of order 16% per qubit before the classical bound is restored; this follows directly from the Horodecki and Werner criteria and is not a new bound. A gap-scaling trade-off follows if the thermal quasiparticle poisoning rate is exponentially suppressed by the gap (Arrhenius factor $e^{-\Delta/k_B T}$), as expected in thermal equilibrium with a spectral gap; for Majorana-based devices, Ref. [28] analyzes the realistic sources, nonequilibrium above-gap quasiparticles and

equilibrium subgap states, and we adopt the simple form for $\nu = 12/5$ by analogy. The universal $\nu = 12/5$ Fibonacci candidate, with a measured activation gap of order tens of mK [30], would then be thermally more fragile than the $\nu = 5/2$ Ising state, whose gap is comparatively robust but which is not computationally universal. A nonequilibrium quasiparticle floor can exceed the equilibrium poisoning rate by orders of magnitude in realistic samples and is decoupled from temperature, so the limitation is not purely thermal and does not vanish as $T \rightarrow 0$. Two further factors pull in opposite directions: at $\nu = 12/5$, Landau-level mixing favors the $k = 3$ Read–Rezayi phase over a competing re-entrant integer quantum Hall phase [29], while fractional quantum Hall states at nearby fillings, observed in the first excited Landau level [33], could add poisoning channels. We do not quote a specific operation-time bound: non-Abelian braiding at $\nu = 12/5$ has not been experimentally demonstrated, so any timescale estimate would be conditional on an identification— $\nu = 12/5$ as the Fibonacci-anyon $k = 3$ Read–Rezayi state—that is numerically supported but not experimentally confirmed [29, 31, 32]. Measurement imperfections beyond this poisoning model are not addressed here.

(ii) The sector phase δ entering via $R_\tau \rightarrow R_\tau \cdot e^{i\delta}$ is a deformation of the representation; for generic $\delta \neq 0$ it does not give a braid-group representation (App. C).

(iii) The fraction of sequences above 2 depends on the evaluation (Sec. V): 91.8% under the four-dimensional projection, which can itself raise product states above 2, and 42.8% for the native five-dimensional Horodecki-type score.

(iv) The simulated autocorrelation (Sec. VI) is quantitatively setup-dependent ($\text{ACF}(1) \in [0.31, 0.36]$ across the four deposited setups), though qualitatively robust ($\text{ACF} > 0$); its transfer to single-shot experimental outcomes is not computed.

(v) In the five-dimensional encoding, a Fourier fit with four harmonics describes $S(\delta)$ at $L = 3$ almost completely ($R^2 = 0.997$) but leaves part of the structure at $L = 8$ and 12 unexplained ($R^2 = 0.837$ and 0.914; App. B).

E. Comparison with the independent fusion-sector construction of Xu, Ye, and You

Independently, Xu, Ye, and You [20] have studied CHSH correlations on triplets of the Fibonacci-anyon state $|\tau, \tau; 1\rangle$ using a graded three-dimensional fusion-sector Hilbert space $\mathcal{H}_1 \oplus \mathcal{H}_\tau$ with measurement operators of the form $A_\alpha = \text{diag}(1, [\cos \alpha, \sin \alpha], [\sin \alpha, -\cos \alpha])$ (their Eq. (23)). Their construction reaches $|S|_{\max} = (4\sqrt{2}\varphi + 2)/\varphi^3 \approx 2.633$ with superactivation as the underlying phenomenon: a single copy of $|\tau, \tau; 1\rangle$ admits a local hidden-variable model, while three copies activate Bell nonlocality. Their setup is bipartite under the graded tensor product, with A_α block-diagonal across $\mathcal{H}_1$

and $\mathcal{H}_\tau$. Tsirelson's bound $2\sqrt{2}$ applies in their setting; the value 2.633 is sub-Tsirelson due to nonuniform Schmidt weights of the three-copy state (weight $2/\varphi^2 \approx 0.764$ on $\mathcal{H}_\tau \otimes \mathcal{H}_\tau$).

The present paper's d_{1b} four-dimensional encoding, by contrast, operates on a single copy of an entangled two-qubit circuit state with concurrence $C = 0.988$, reaching $|S| = 2.811$ at $\delta/\pi = 1.76$ (at $\delta = 0$ the maximum over all words, $|S| = 2.733$, belongs to a different word); this is the encoding-level two-qubit Horodecki value $2\sqrt{1+C^2}$ for this concurrence. The two constructions address different questions (multi-copy activation versus single-copy circuit-model violation) and produce different but compatible values: both respect Tsirelson's bound, and the difference reflects distinct Schmidt structures of the underlying states. The value $|S| = 3.406$ of the companion paper [2] belongs to a nonbipartite construction that violates setting-locality and is therefore not subject to Tsirelson's bound.

F. Outlook

The present enumeration is bounded at $L = 12$ by the 3^L scaling; other anyon models (Ising, $SU(2)_k$ for $k > 3$) and experimental autocorrelation data are not addressed here. Whether the Kitaev honeycomb model [14], which offers a lattice setting in which both classical and quantum limits are accessible, can bridge the present work with the $\mathbb{Z}_2$ results of Ref. [1] remains an open question. A further open question, suggested by the Schmidt-rank-3 structure of the Xu, Ye, You construction [20], is whether a single-copy Schmidt-rank-3 state in the five-dimensional fusion space could be prepared by a different braiding protocol, and which CHSH values such a state would yield compared with the present results: $|S| = 2.733$ with the standard Fibonacci data and $|S| = 2.811$ at the sector-phase optimum $\delta/\pi = 1.76$ in the two-generator circuit model, and $|S| = 2.824$ in the three-generator encoding under four-dimensional projection at $L = 8$.

VIII. CONCLUSION

We have presented a complete sequence-by-sequence enumeration of CHSH-form values across all Fibonacci anyon braiding words of length $L = 3$ –12 (two generators) and $L = 2$ –9 (three generators, with the $\delta = 0$ maxima also at $L = 10$ and 12) in two complementary encodings, in the broader landscape of qualitative Bell-test constructions on Fibonacci/Ising and Majorana platforms [15–19] and the independently analyzed three-copy fusion-sector construction of Xu, Ye, and You [20]; a companion Leggett–Garg-inequality letter using the same Fibonacci F - and R -matrices, with K_3 values reaching 99.998% of the Lüders bound $3/2$, is reported in Ref. [22].

Braiding alone does not entangle the logical qubits of the two-generator encoding: on the fusion code all 8190 braid words of length $L = 1$ –12 give $|S| = 2$ exactly, while the two-qubit circuit built from the same data exceeds 2. In the three-generator encoding, values above 2 are common: 91.8% of the sequences exceed 2 under computational-subspace projection at $L = 9$ (including 510 words without σ_3 that do so only through the projection), and 42.8% under the native five-dimensional score, where a further 28.5% lie exactly at the classical value.

In the d_{1b} circuit model the $L = 12$ maximum is $|S| = 2.733$ (96.6% of the Tsirelson bound $2\sqrt{2}$, $C = 0.932$) with the standard Fibonacci data ($\delta = 0$) and $|S| = 2.811$ (99.4%, $C = 0.988$) at the sector-phase optimum, while the three-generator encoding under four-dimensional projection reaches $|S| = 2.824$ (99.84%) at $L = 8$.

In the three-generator encoding, removing the cross-bipartition generator σ_3 leaves only gates local to Alice or Bob and, from the product input state, collapses entanglement to zero exactly (with $|11\rangle$ assigned to $|4\rangle$), independent of sequence length and sector phase; in the d_{1b} circuit model, removing either of its two gates likewise gives $C = 0$.

Sequential braiding on a shared anyon register produces positive lag-1 autocorrelation of the simulated CHSH value ($ACF(1) = +0.35$ at 11.22σ), a signature of state continuity: the state is carried from step to step, and each step applies a single generator. Whether single-run outcomes in an experiment, with a measurement in every run, retain a positive autocorrelation is not computed here (Sec. VI).

We emphasize that the CHSH values reported here, while in excess of the classical bound, should not be interpreted as Bell violations in the EPR sense (cf. Sec. II E): the underlying Fibonacci fusion space does not factorize as a tensor product of local Alice and Bob subsystems, and the classical-bound excess is more precisely characterized as a signature of nonseparability in fusion space.

These findings indicate that the available braiding operations determine whether CHSH-form values above 2 are accessible on the fusion space.

NOTATION

Symbols used in this paper.

Symbol	Meaning
L	braiding sequence length
$ S $	magnitude of the CHSH functional
C	concurrence of the two-qubit state
AM, MB	the two gates of the d_{1b} circuit model: AM acts on Alice's qubit, MB on Bob's qubit conditioned on Alice's; they are not the braid generators σ_2, σ_4
$\sigma_2, \sigma_3, \sigma_4$	braid generators of the three-generator encoding; σ_2 and σ_4 alone act on the fusion code of the two-generator encoding

d_{1b}	the two-generator encoding of six Fibonacci anyons
δ	sector phase, a deformation parameter introduced through $R_\tau \rightarrow R_\tau e^{i\delta}$; only $\delta = 0$ is the standard theory
φ	golden ratio, $(1 + \sqrt{5})/2$

DATA AND CODE AVAILABILITY

The deterministic scripts and JSON output files that validate the results reported in this paper are archived on Zenodo under the concept DOI [10.5281/zenodo.19601352](https://doi.org/10.5281/zenodo.19601352) (which always resolves to the latest version). Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License 2.0 (see [LICENSE](#) and [LICENSE-CODE](#)). The archive includes the three-generator (5D) landscape validation and its leakage-norm reanchor (`landscape.validation.py`, `n2.reanchor.py`); the Protocol A projection audit, including the first $\delta = 0$ maximizers of the three-generator encoding for $L = 3, 6, 8, 10, 12$ over all 3^L words (`d10.protocol.a.projection.audit.py`); a closed-form sector-phase sweep validation covering all rows of Table I and the three-generator $\delta = 0$ maximizers at $L = 3, 6, 8, 10, 12$ (`d4.s.delta.sweep.validation.py`, run after `d10`, reproducing `s.delta.sweeps.json` and the printed table values); the σ_3 -blocking validation, including the exact σ_3 -block $C = 0$ check (`d5.topology.switch.validation.py`, reproducing `topology.switch.results.json`); the braiding-only computation on the leakage-free fusion code and the Yang–Baxter residua and distant commutator of the generators used here (`d9.fusion.code.braiding.only.py`); the Fourier fits of Appendix B (`d11.fourier.fit.sweeps.py`); the sequential-braiding autocorrelation of Table IV (`d6.acf.prediction.sequential.braiding.py`); the seeded Pillar-2 null test and initialization-robustness evaluation of Sec. VI (`d8.acf.null.robustness.py`); an independent null-control reanalysis of the Hensen *et al.* (2015) loophole-free Bell data, including the underlying public dataset (`d7.acf.hensen2015.null.control.py`); and the scripts that draw Figs. 1, 3 and 4 (`figures/make.1b.figs.py`) and Fig. 2 (`figures/make.1b.fig4.map.py`) from the deposited JSON files. The two-generator sequences of Table I come from the full enumeration deposited with the companion record [1]. The checks of the remaining generator pairs in Appendix C and the $\mathbb{Z}_5$ phase spectrum of Δ^2 were made with two further scripts that are not part of this record.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though

this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI’s.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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Appendix A: Braid-generator matrices

The five braid generators $\sigma_1, \dots, \sigma_5$ for six Fibonacci anyons in the basis (1) are constructed as follows. For each generator σ_k , the anyon pair (τ_k, τ_{k+1}) is braided via the F -move/ R -phase/ F -move sequence on the relevant two-dimensional subspace of the fusion tree. The remaining basis states acquire diagonal R -phases. Explicit 5×5 matrices are provided in the supplementary code archived with the Zenodo deposit.

The key structural properties:

- σ_2 mixes states within the $x_1 \in \{1, \tau\}$ subspace (Alice’s qubit), preserving x_3 .
- σ_4 mixes states within the $x_3 \in \{1, \tau\}$ subspace (Bob’s qubit), preserving x_1 .
- σ_3 re-fuses the intermediate label x_2 , coupling both branches of the fusion tree.
- σ_1 and σ_5 act as diagonal phases on x_1 and x_3 respectively.

Appendix B: Fourier structure of $S(\delta)$

The CHSH value as a function of sector phase δ exhibits encoding-dependent harmonic structure. For the Ising anyon model, the functional form is exactly $S(\delta) = 2\sqrt{2}\cos^2(\delta/2)$, containing only the fundamental harmonic $n = 1$. This idealized Ising result assumes access to a nonstabilizer (“magic”) single-qubit operation beyond the topologically protected Clifford gate set; purely stabilizer operations on Ising anyons cannot violate the CHSH inequality [42]. For Fibonacci anyons we fit the deposited five-dimensional sweeps of the first $\delta = 0$ maximizers at $L = 3, 8$ and 12 (Fig. 1) by least squares with $N = 1, \dots, 6$ harmonics,

$$S(\delta_k) \approx a_0 + \sum_{n=1}^N (a_n \cos n\delta_k + b_n \sin n\delta_k), \quad \delta_k = 2\pi k/50, \quad (\text{B1})$$

(`d11_fourier_fit_sweeps.py`; two solution routes agree to 5×10^{-16}). With $N = 4$ the coefficient of determination is $R^2 = 0.997, 0.837$ and 0.914 at $L = 3, 8$ and 12 ; with $N = 6$ it is $1.000, 0.919$ and 0.952 . For every $N \geq 2$ the largest harmonic is $n = 2$, with amplitude $\sqrt{a_2^2 + b_2^2} = 0.388, 0.280$ and 0.273 , and the constant term is the mean of the curve ($a_0 = 2.280, 2.243, 2.315$). For Fibonacci, generic $\delta \neq 0$ is a deformation that does not preserve the braid relations (App. C), so we attach no physical meaning to the harmonic content of these fits; they describe the numerical curves only.

Appendix C: Verification of the five-dimensional representation under sector-phase modulation

The qualitative nonseparability and contextuality framing of Sec. II E and Sec. II D is supported by an independent numerical check of the braid-group relations for the full five-state generators $\sigma_1, \dots, \sigma_5$ across a range of sector phases δ . At $\delta = 0$ the standard Fibonacci modular tensor category is satisfied to floating-point precision (Yang–Baxter residuum below 10^{-15} , distant commutators identically zero). Under the sector-phase modulation $R_\tau \mapsto R_\tau \cdot e^{i\delta}$ introduced in Sec. II D, Yang–Baxter deforms uniformly across all four neighboring generator pairs while distant commutators $[\sigma_i, \sigma_j] = 0$ for $|i - j| \geq 2$ remain identically zero throughout. Table V summarizes the verification. The sector phase δ is thus a free deformation knob rather than a physical parameter: generic $\delta \neq 0$ breaks the Yang–Baxter/hexagon consistency of the representation (Table V), and only $\delta = 0$ recovers the standard Fibonacci modular tensor category. This expected discreteness — rather than a continuum of consistent deformations — is consistent with the general Ocneanu-rigidity result for fusion categories [27], which here rests on the paper-

internal numerics of Table V rather than on an independent categorical classification of this specific representation.

On the 50-point grid the braid relations hold exactly only at $\delta/\pi = 0, 0.6$ and 1.2 : at $\delta = 3\pi/5$ all generators become multiples of the identity, and at $\delta = 6\pi/5$ they are the complex conjugates of the $\delta = 0$ generators up to a global phase. Everywhere else on the grid the Yang–Baxter residuum is at least 0.196 (`d9_fusion_code_braiding_only.py`). More generally, complex conjugation maps the braid generators at δ to those at $6\pi/5 - \delta$ up to a global phase, and concurrence is invariant under complex conjugation; numerically, every $|S|(\delta)$ sweep of this paper, in both encodings, is mirror-symmetric about $\delta = 3\pi/5$ and $8\pi/5$ (deviation below 10^{-14}), which is why the d_{1b} optimum appears twice, at $\delta/\pi = 1.44$ and 1.76 .

TABLE V. Algebraic verification of the five-state representation under sector-phase modulation. The Yang–Baxter residuum $\|\sigma_i \sigma_{i+1} \sigma_i - \sigma_{i+1} \sigma_i \sigma_{i+1}\|_F$ is identical across all four neighboring pairs and is shown in column 2. The distant commutator $\|[\sigma_i, \sigma_j]\|_F$ for $|i - j| \geq 2$ vanishes for every tested δ , as the block structure of the generators requires; the deposited check of $[\sigma_2, \sigma_4]$, the pair used in this paper, finds at most 8×10^{-17} , i.e. floating-point rounding.

δ/π	YB residuum (5D)	distant commutator (5D)
0.000	$< 10^{-15}$	$< 10^{-16}$
0.100	0.525	$< 10^{-16}$
0.250	0.958	$< 10^{-16}$
0.500	0.474	$< 10^{-16}$
1.000	0.873	$< 10^{-16}$
1.191	0.052	$< 10^{-16}$
1.500	1.523	$< 10^{-16}$

The intact $\delta = 0$ representation confirms that the non-commuting d_{1b} gates belong to the circuit model, not to the braid representation (Sec. II D): in the fusion code, σ_2 and σ_4 commute exactly. The exact preservation of distant commutativity under δ -modulation is structural: the modulation acts locally on the R_τ eigenvalue and does not couple block- or diagonally-separated generators. The Yang–Baxter residua of the generators used in this paper and their distant commutator are recomputed by the deposited script `d9_fusion_code_braiding_only.py`, which finds the commutator below 10^{-16} at every tested δ , as the structural argument requires. The remaining pairs, the $\mathbb{Z}_5$ phase spectrum of Δ^2 from the Birman formula and the d_{1b} spectator-filter algebra-breaks were computed with two further scripts (`d2_delta_phase_consistency.py`, `d3_5d_braid_algebra_consistency.py`) that are not part of this record.

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